Assignment 4

1. Problem Statement: Design a comprehensive course registration portal combining multiple collection types for optimal data handling, prerequisites tracking, and waitlist management.

Use a HashMap <String, Course> for fast course lookups by course code. Each Course object must contain a HashSet of unique enrolled student IDs, an ArrayList for ordered prerequisite codes, and a LinkedList acting as a first-come, first-served waitlist queue. Handle registration requests with capacity checks and automated waitlist promotion logic.

2. Problem Statement: Build a media player queue engine that supports fast insertions at arbitrary positions, recent history tracking, and bidirectional navigation.

Utilize a Linked List<Song> to manage playback queues. Implement playlist features that allow adding songs to the front (priority queueing), removing duplicate song entries efficiently, and using List Iterator to traverse the playlist both forward and backward without index-based performance bottlenecks.